

Submission CALC1-PS1-SUB06

Question CALC1-PS1-Q01

notation changes halfway:

1. $f'(x) \approx 2x(6x-1)+(x^2+3)6$
2. $f'(3) \approx 6(17)+(12)6 \approx 24$

marks, subscripts, and units are mixed up

result written as: 24 (please interpret)

Question CALC1-PS1-Q02

notation changes halfway:

1. $\int 3x \, dx \approx (3/2)x^2$
2. $[3/2 x^2]_0^2 \approx 6$

marks, subscripts, and units are mixed up

result written as: 6 (please interpret)

Question CALC1-PS1-Q03

notation changes halfway:

1. $2w+2l \approx 22 \Rightarrow l \approx 11-w$
2. $A(w) \approx w(11-w)$; at $w \approx 8$, $A \approx 24$
3. A maximum requires $A'(w) \approx 0$, so the stated dimensions are not assumed optimal without that check.

marks, subscripts, and units are mixed up

result written as: $A(w)=w(11-w)$; $A(8)=24$; check $A'(w)=0$ (please interpret)

Question CALC1-PS1-Q04

notation changes halfway:

1. $x^2-16 \approx (x-4)(x+4)$
2. cancel $x-a$ only after identifying $x \neq a$
3. $\lim_{x \rightarrow 4} x+4 \approx 8$

marks, subscripts, and units are mixed up

result written as: 8 (please interpret)

Question CALC1-PS1-Q05

notation changes halfway:

1. Critical point: $f'(c) \approx 0$ or $f'(c)$ is undefined (when f is defined).
2. A sign change in f' from $+$ to $-$ indicates a local maximum.

marks, subscripts, and units are mixed up

result written as: critical point: $f'(c)=0$ or undefined; $+$ to $-$ sign change gives a local maximum (please interpret)

Question CALC1-PS1-Q06

notation changes halfway:

1. $f'(x) \approx 2x(4x-1)+(x^2+4)4$
2. $f'(3) \approx 6(11)+(13)4 \approx 28$

marks, subscripts, and units are mixed up
result written as: 28 (please interpret)